Project: Worked examples
Title: Cylinder and actuator circuit
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1. Scope and method

A double-acting hydraulic cylinder in its circuit: the bore, rod and annulus areas; the force and speed on the extend and the retract stroke with mechanical and volumetric efficiency and an optional back pressure, or with the regenerative extend connection; the swept volume and the cycle time; the pump delivery, shaft power, torque and speed; a hydraulic motor's torque; the velocity in the suction, pressure and return lines against your own limits and the smallest listed hose or pipe bore that meets them; a sharp orifice's flow for flow-control screening; and a screening check of the rod against buckling (Euler above the transition slenderness, Johnson parabola below it). screening only; ISO/TS 13725 method not implemented. Not a cylinder design to a standard: the limits, the hose sizes, the mounting factor and the material properties are your records.
Areas: $A_b = \pi \cdot D^2/4$, $A_r = \pi \cdot d_r^2/4$, $A_a = A_b - A_r$, area ratio $\varphi = A_b/A_a$ Force: $F_{ext} = (p \cdot A_b - p_{back} \cdot A_a) \cdot \eta_m$, $F_{ret} = (p \cdot A_a - p_{back} \cdot A_b) \cdot \eta_m$; regenerative extend $F_{ext} = p \cdot A_r \cdot \eta_m$ Speed and time: $v_{ext} = Q \cdot \eta_{vc}/A_b$ (regenerative: A_r), $v_{ret} = Q \cdot \eta_{vc}/A_a$; $t = stroke/v$; $t_{cycle} = t_{ext} + t_{ret} + t_{dwell}$ Required speed (optional): when v_{req} is given it gives the flow, $Q = v_{req} \cdot A_b/\eta_{vc}$ (regenerative: A_r), and replaces the entered Q; the flow must lie within 0.1 to 2000 L/min Swept volume: $\Delta V_{ext} = A_b \cdot stroke$ (regenerative: $A_r \cdot stroke$), $\Delta V_{ret} = A_a \cdot stroke$; ΔV = the larger Pump: $Q_p = V_g \cdot n \cdot \eta_{vp}$; $P = p \cdot Q_p/\eta_{tp}$; $T = P/\omega$; displacement needed $V_{g,req} = Q/(n \cdot \eta_{vp})$ Hydraulic motor: $T = \Delta p \cdot V_g/(2\pi) \cdot \eta_{hm}$ (V_g per revolution) Lines: $v = Q_{line}/A_{line}$; suction $Q_{suc} = \max(Q, V_g \cdot n)$ (the geometric delivery), pressure/cap-port line $Q_{prs} = Q_{line} \cdot A_b/A_a$ on the retract stroke (regenerative extend: $Q_{line} \cdot A_b/A_r$ if larger), return $Q_{rtn} = Q_{line} \cdot A_b/A_a$, with $Q_{line} = \max(Q, Q_p)$; smallest bore $d = \sqrt{(4Q/(\pi \cdot v_{max}))}$, recommended = the smallest listed size not below it Orifice: $Q = C_d \cdot A_o \cdot \sqrt{(2\Delta p/\rho)}$; bore for the circuit flow $d_o = \sqrt{(4Q/(\pi \cdot C_d \cdot \sqrt{(2\Delta p/\rho)}))}$ Rod buckling (screening): $L_{eff} = K \cdot (stroke + L_{add})$, $r = d_r/4$, $\lambda = L_{eff}/r$, $\lambda_c = \pi \cdot \sqrt{(2E/f_y)}$; Euler $F_{cr} = \pi^2 \cdot E \cdot I/L_{eff}^2$ for $\lambda > \lambda_c$, Johnson $F_{cr} = A_r \cdot f_y \cdot [1 - f_y \cdot \lambda^2/(4\pi^2 E)]$ for $\lambda \le \lambda_c$; load $F_b = \max(F_{push}, F_{stall})$ with $F_{stall} = p \cdot A_b$ (the stalled cylinder at the supply pressure: no flow, so no back pressure and no friction loss; in the regenerative extend too, since the standard connection can always be selected); utilisation $u = F_b \cdot S_f/F_{cr}$
Ref: ISO/TS 13725 — Cylinder buckling load: method not implemented, screening only
Ref: Esposito, Fluid Power with Applications Cylinder and pump chapters — Force, speed, flow and power relations, closed form

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mode	Extend connection	Standard		assumption: Standard connection: the ram needs its full bore force for the press stroke.	any
p	Supply pressure at the cylinder and the pump (gauge)	180	bar	spec: Hydraulic specification HS-31: relief valve set 180 bar at the ram	1 … 700 bar

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
p_back	Back pressure on the discharge side (standard connection)	0	bar	assumption: Return line direct to tank; the return filter drop is neglected for this example.	0 … 700 bar
Q	Flow into the cylinder (blank when a required extend speed gives it)	60	L/min	spec: Hydraulic specification HS-31: ram flow 60 L/min	0.1 … 2000 L/min
rho	Fluid density	870	kg/m^3	reference: fluid data sheet: mineral oil, density at 40 °C, to be taken from the oil supplier's data	700 … 1300 kg/m^3
F_push	Load to push on the extend stroke (compresses the rod)	150	kN	spec: Hydraulic specification HS-31: pressing force 150 kN	0 … 15000 kN
F_pull	Load to pull on the retract stroke	60	kN	spec: Hydraulic specification HS-31: return stroke load 60 kN	0 … 15000 kN
D	Bore	125	mm	drawing: DWG-4310 rev B	10 … 500 mm
d_r	Rod diameter	90	mm	drawing: DWG-4310 rev B	4 … 450 mm
stroke	Stroke	800	mm	drawing: DWG-4310 rev B	1 … 5000 mm
eta_m	Mechanical efficiency of the cylinder (seal and guide friction)	0.95		assumption: Typical seal and guide friction for a new cylinder; to be confirmed against the cylinder maker's figure.	0.5 … 1
eta_vc	Volumetric efficiency of the cylinder (leakage past the piston)	0.98		assumption: Allowance for leakage past the piston; to be confirmed against the cylinder maker's figure.	0.5 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
t_dwell	Dwell time added to each cycle	0	s	assumption: Cycle time of the strokes only; dwell added by the machine controller.	0 … 3600 s
K_m	Effective-length factor K of the mounting (L_eff = K·L)	1		reference: record: cylinder mounting (clevis at both ends, pinned), to be taken from the cylinder maker's mounting guidance	0.5 … 2.5
L_add	Length added to the stroke for the buckling length (mounting and rod end beyond the stroke)	150	mm	drawing: DWG-4310 rev B	0 … 5000 mm
E	Young's modulus of the rod	210	GPa	reference: record: rod material (chromed carbon steel), to be taken from the material library	50 … 250 GPa
f_y	Yield strength of the rod	500	MPa	reference: record: rod material (chromed carbon steel), to be taken from the rod maker's certificate	100 … 1500 MPa
S_f	Safety factor on the critical load	3.5		assumption: Safety factor chosen for this example; the design reference sets the real one.	1 … 20
V_g	Pump displacement per revolution (leave blank to skip the pump delivery check)	45	cm^3/rev	spec: Hydraulic specification HS-31: fixed-displacement pump 45 cm³/rev	0.1 … 1000 cm^3/rev
n_p	Pump speed	1450	rpm	spec: Hydraulic specification HS-31: 4-pole motor, 1450 rpm	50 … 6000 rpm
eta_vp	Volumetric efficiency of the pump	0.95		assumption: Pump volumetric efficiency at 180 bar; to be taken from the pump data sheet.	0.5 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
eta_tp	Total efficiency of the pump (volumetric × hydromechanical)	0.88		assumption: Pump total efficiency at 180 bar; to be taken from the pump data sheet.	0.5 … 1
eta_hm	Hydromechanical efficiency of the motor	0.9		default	0.5 … 1
d_suc	Suction line inside diameter	50	mm	drawing: DWG-4311 rev A	3 … 200 mm
d_prs	Pressure line inside diameter (to the cylinder cap port; judged at the larger flow of the two strokes)	25	mm	drawing: DWG-4311 rev A	3 … 200 mm
d_ret	Return line inside diameter	32	mm	drawing: DWG-4311 rev A	3 … 200 mm
v_suc_max	Suction line velocity limit (your record; the default is a placeholder, not a code value)	1.2	m/s	reference: record: line velocity limits, a placeholder to be replaced from the design reference	0.1 … 20 m/s
v_prs_max	Pressure line velocity limit (your record; the default is a placeholder, not a code value)	5	m/s	reference: record: line velocity limits, a placeholder to be replaced from the design reference	0.1 … 20 m/s
v_ret_max	Return line velocity limit (your record; the default is a placeholder, not a code value)	3	m/s	reference: record: line velocity limits, a placeholder to be replaced from the design reference	0.1 … 20 m/s
hoses	Available hose or pipe inside diameters (from your supplier's range)	7 row(s)		reference: record: illustrative hose size list invented for this example, to be replaced by the supplier's range	any
hoses[1]		ID = 16 mm			
hoses[2]		ID = 19 mm			
hoses[3]		ID = 25 mm			
hoses[4]		ID = 32 mm			

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
hoses[5]		ID = 38 mm			
hoses[6]		ID = 51 mm			
hoses[7]		ID = 63 mm			
C_d	Orifice discharge coefficient	0.62		default	0.2 … 1

3. Assumptions

A1. The supply pressure p is the gauge pressure at the cylinder port and, for the pump, at the pump outlet; line losses are not subtracted (the line losses of a pipe run come from the pipe calculation). The force is the pressure on the area less the back pressure on the other side, times the mechanical efficiency. (F = p·A·η_m, closed form.)
A2. The speed is the flow into the cylinder times the volumetric efficiency over the area, in steady motion: no acceleration, cushioning or compressibility. The cycle time is the two strokes plus the dwell. (v = Q·η_vc/A.)
A3. The pump power is the supply pressure times the pump's delivery over the total efficiency: when the pump delivers more than the circuit takes, the excess is taken to go over the relief valve at the supply pressure, which is the conservative case for the drive. The shaft torque is the power over the angular speed. (P = p·Q_p/η_tp.)
A4. The line flows use the pump delivery when it is larger than the circuit flow: the suction line the geometric delivery V_g·n, the pressure line and the return line Q_line = max(Q, Q_p), which carries no volumetric efficiency while the speeds use Q·η_vc (the flows are the conservative side of the speeds). The line to the cylinder cap port is judged at the larger flow of the two strokes: on the retract stroke it carries the cap-side discharge Q_line·A_b/A_a (the same flow as the return line), and in the regenerative extend Q_line·A_b/A_r if that is larger. The velocity limits are your records; the sheet's defaults are placeholders, not values from a standard. (Continuity.)
A5. The orifice relation is the sharp-edged turbulent one, Q = C_d·A·√(2Δp/ρ), with your discharge coefficient; it ignores viscosity and cavitation and is a screening of a flow-control orifice, not a valve selection. (Orifice equation, closed form.)
A6. The rod is treated as a solid round column of its own diameter with the effective-length factor of the mounting; the tube, the piston and the rod-end stiffness are not modelled. The rod is judged at the larger of the push load you enter and the stall thrust p·A_b (a stalled, obstructed or relief-limited cylinder loads the rod with the full pressure force on the bore: no flow, so no back pressure, and the seal friction does not relieve it); no pressure-limiting device is credited, and the regenerative extend is judged at the same stall thrust because the standard connection can always be selected. The mounting factor K_m defaults to 1 (pinned at both ends): a fixed-free mounting is K_m = 2, and the factor and the safety factor S_f are your records. screening only; ISO/TS 13725 method not implemented. (Euler and Johnson column relations, closed form.)
A7. mode (Extend connection) is taken as Standard. (Standard connection: the ram needs its full bore force for the press stroke.)
A8. p_back (Back pressure on the discharge side (standard connection)) is taken as 0 bar. (Return line direct to tank; the return filter drop is neglected for this example.)
A9. eta_m (Mechanical efficiency of the cylinder (seal and guide friction)) is taken as 0.95. (Typical seal and guide friction for a new cylinder; to be confirmed against the cylinder maker's figure.)
A10. eta_vc (Volumetric efficiency of the cylinder (leakage past the piston)) is taken as 0.98. (Allowance for leakage past the piston; to be confirmed against the cylinder maker's figure.)
A11. t_dwell (Dwell time added to each cycle) is taken as 0 s. (Cycle time of the strokes only; dwell added by the machine controller.)

A12. S_f (Safety factor on the critical load) is taken as 3.5. (Safety factor chosen for this example; the design reference sets the real one.)
A13. eta_vp (Volumetric efficiency of the pump) is taken as 0.95. (Pump volumetric efficiency at 180 bar; to be taken from the pump data sheet.)
A14. eta_tp (Total efficiency of the pump (volumetric × hydromechanical)) is taken as 0.88. (Pump total efficiency at 180 bar; to be taken from the pump data sheet.)
A15. Standard connection: the extend force is the pressure on the bore area less the back pressure on the annulus area; the retract force the pressure on the annulus less the back pressure on the bore. (Continuity and force balance.)

4. Working

```
[28] A_b = pi * D^2 / 4
= π × (125 mm)^2 / 4
= 122.72 cm²
```
Bore (cap-side) area
```
[29] A_r = pi * d_r^2 / 4
= π × (90 mm)^2 / 4
= 63.617 cm²
```
Rod area
```
[30] A_a = A_b - A_r
= 122.72 cm² - 63.617 cm²
= 59.101 cm²
```
Annulus (rod-side) area
```
[31] phi = A_b / A_a
= 122.72 cm² / 59.101 cm²
= 2.0764
```
Area ratio A_b/A_a
```
[33] F_ext = (p * A_b - p_back * A_a) * eta_m
= (180 bar × 122.72 cm² - 0 bar × 59.101 cm²) × 0.95
= 209.85 kN
```
Extend force
```
[34] v_ext = Q * eta_vc / A_b
= 60 L/min × 0.98 / 122.72 cm²
= 79.858 mm/s
```
Extend speed
```
[35] dV_ext = A_b * stroke
= 122.72 cm² × 800 mm
= 9.8175 L
```
Liquid taken from the supply for the extend stroke
```
[36] F_ret = (p * A_a - p_back * A_b) * eta_m
= (180 bar × 59.101 cm² - 0 bar × 122.72 cm²) × 0.95
= 101.06 kN
```
Retract force
```
[37] v_ret = Q * eta_vc / A_a
= 60 L/min × 0.98 / 59.101 cm²
= 165.82 mm/s
```
Retract speed
```
[38] t_ext = stroke / v_ext
= 800 mm / 79.858 mm/s
= 10.018 s
```
Extend stroke time
```
[39] t_ret = stroke / v_ret
= 800 mm / 165.82 mm/s
= 4.8246 s
```

Retract stroke time

[40] t_cycle = t_ext + t_ret + t_dwell
= 10.018 s + 4.8246 s + 0 s
= 14.842 s

Cycle time

[41] dV_ret = A_a * stroke
= 59.101 cm² × 800 mm
= 4.7281 L

Liquid taken from the supply for the retract stroke

[42] dV = max(dV_ext, dV_ret)
= max(9.8175 L, 4.7281 L)
= 9.8175 L

Swept volume of the larger stroke

[43] Q_p = V_g * n_p * eta_vp
= 45 cm³/rev × 1450 rpm × 0.95
= 61.988 L/min

Pump delivery

[44] Vg_req = Q / (n_p * eta_vp)
= 60 L/min / (1450 rpm × 0.95)
= 43.557 cm³/rev

Pump displacement the circuit flow needs at this speed

[45] u_pump = Q / Q_p
= 60 L/min / 61.988 L/min
= 0.96794

Circuit flow over the pump delivery

[46] Q_pump = Q_p
= 61.988 L/min
= 61.988 L/min

Flow the pump delivers at the supply pressure

[47] P_hyd = p * Q_pump
= 180 bar × 61.988 L/min
= 18.596 kW

Hydraulic power at the pump outlet

[48] P_pump = P_hyd / eta_tp
= 18.596 kW / 0.88
= 21.132 kW

Pump shaft power

[49] T_shaft = P_pump / n_p
= 21.132 kW / 1450 rpm
= 139.17 N·m

Pump shaft torque P / ω

[50] n_shaft = n_p
= 1450 rpm
= 1450 rpm

Pump shaft speed

[51] Q_line = max(Q, Q_p)
= max(60 L/min, 61.988 L/min)
= 61.988 L/min

Flow in the pressure and return lines before the area ratio (the larger of the circuit flow and the pump delivery)

[52] Q_suc = max(Q, V_g * n_p)
= max(60 L/min, 45 cm³/rev × 1450 rpm)
= 65.25 L/min

Flow in the suction line (the larger of the circuit flow and the geometric pump delivery V_g·n)

[53] Q_prs = Q_line * A_b / A_a
= 61.988 L/min × 122.72 cm² / 59.101 cm²
= 128.71 L/min
Flow in the line to the cap port, worst stroke (retract: the cap-side discharge Q·A_b/A_a)
[54] Q_rtn = Q_line * A_b / A_a
= 61.988 L/min × 122.72 cm² / 59.101 cm²
= 128.71 L/min
Flow in the return line on the retract stroke (cap-side discharge)
[55] v_suc = Q_suc / (pi * d_suc^2 / 4)
= 65.25 L/min / (π × (50 mm)^2 / 4)
= 0.55386 m/s
Velocity in the suction line
[56] u_suc = v_suc / v_suc_max
= 0.55386 m/s / 1.2 m/s
= 0.46155
Velocity in the suction line over its limit
[57] d_min_suc = sqrt(4 * Q_suc / (pi * v_suc_max))
= sqrt(4 × 65.25 L/min / (π × 1.2 m/s))
= 33.969 mm
Smallest suction line bore for the velocity limit
[58] d_rec_suc = smallest listed size not below the minimum bore
= d_min = 33.969 mm, 7 sizes listed
= 38 mm
Recommended suction line bore
[59] v_prs = Q_prs / (pi * d_prs^2 / 4)
= 128.71 L/min / (π × (25 mm)^2 / 4)
= 4.3702 m/s
Velocity in the pressure line
[60] u_prs = v_prs / v_prs_max
= 4.3702 m/s / 5 m/s
= 0.87403
Velocity in the pressure line over its limit
[61] d_min_prs = sqrt(4 * Q_prs / (pi * v_prs_max))
= sqrt(4 × 128.71 L/min / (π × 5 m/s))
= 23.372 mm
Smallest pressure line bore for the velocity limit
[62] d_rec_prs = smallest listed size not below the minimum bore
= d_min = 23.372 mm, 7 sizes listed
= 25 mm
Recommended pressure line bore
[63] v_rtn = Q_rtn / (pi * d_ret^2 / 4)
= 128.71 L/min / (π × (32 mm)^2 / 4)
= 2.6673 m/s
Velocity in the return line
[64] u_rtn = v_rtn / v_ret_max
= 2.6673 m/s / 3 m/s
= 0.88911
Velocity in the return line over its limit
[65] d_min_rtn = sqrt(4 * Q_rtn / (pi * v_ret_max))
= sqrt(4 × 128.71 L/min / (π × 3 m/s))
= 30.174 mm
Smallest return line bore for the velocity limit

[66] d_rec_rtn = smallest listed size not below the minimum bore
= d_min = 30.174 mm, 7 sizes listed
= 32 mm
Recommended return line bore
[67] L_b = stroke + L_add
= 800 mm + 150 mm
= 950 mm
Buckling length: stroke plus the length beyond it
[68] L_eff = K_m * L_b
= 1 × 950 mm
= 950 mm
Effective length K·L
[69] I_r = pi * d_r^4 / 64
= π × (90 mm)^4 / 64
= 3220600 mm⁴
Second moment of area of the rod
[70] r_g = d_r / 4
= 90 mm / 4
= 22.5 mm
Radius of gyration of the solid rod
[71] lam = L_eff / r_g
= 950 mm / 22.5 mm
= 42.222
Slenderness ratio
[72] lam_c = pi * sqrt(2 * E / f_y)
= π × sqrt(2 × 210 GPa / 500 MPa)
= 91.052
Transition slenderness between Johnson and Euler
[73] F_cr = A_r * f_y * (1 - f_y * lam^2 / (4 * pi^2 * E))
= 63.617 cm² × 500 MPa × (1 - 500 MPa × 42.222^2 / (4 × π^2 × 210 GPa))
= 2838.9 kN
Critical load, Johnson parabola (stocky rod)
[74] F_allow = F_cr / S_f
= 2838.9 kN / 3.5
= 811.11 kN
Allowable push load: critical load over the safety factor
[75] F_stall = p * A_b
= 180 bar × 122.72 cm²
= 220.89 kN
Stall thrust: the supply pressure on the bore area (no flow, so no back pressure; the friction of the seals does not relieve the rod of a stalled cylinder)
[76] F_buck = max(F_push, F_stall)
= max(150 kN, 220.89 kN)
= 220.89 kN
Rod load for the buckling check: the larger of the push load entered and the stall thrust p·A_b
[77] u_buck = F_buck / F_allow
= 220.89 kN / 811.11 kN
= 0.27234
Buckling utilisation
[78] u_push = F_push / F_ext
= 150 kN / 209.85 kN
= 0.7148
Push load over the extend force

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[79] u_pull = F_pull / F_ret
= 60 kN / 101.06 kN
= 0.59369
```
Pull load over the retract force

5. Results

SYMBOL	VALUE	DESCRIPTION
A_b	122.72 cm²	Bore area
A_r	63.617 cm²	Rod area
A_a	59.101 cm²	Annulus area
phi	2.0764	Area ratio A_b/A_a
F_ext	209.85 kN	Extend force
F_ret	101.06 kN	Retract force
v_ext	79.858 mm/s	Extend speed
v_ret	165.82 mm/s	Retract speed
t_ext	10.018 s	Extend time
t_ret	4.8246 s	Retract time
t_cycle	14.842 s	Cycle time
dV_ext	9.8175 L	Extend swept volume
dV_ret	4.7281 L	Retract swept volume
dV	9.8175 L	Swept volume per stroke (the larger)
u_push	0.7148	Push load over the extend force
u_pull	0.59369	Pull load over the retract force
Q	60 L/min	Flow into the cylinder
p	180 bar	Supply pressure
rho	870 kg/m³	Fluid density
d_prs	25 mm	Pressure line inside diameter
Q_p	61.988 L/min	Pump delivery
Vg_req	43.557 cm³/rev	Pump displacement the circuit needs
u_pump	0.96794	Circuit flow over the pump delivery
P_pump	21.132 kW	Pump shaft power
T_shaft	139.17 N·m	Pump shaft torque
n_shaft	1450 rpm	Pump shaft speed
Q_line	61.988 L/min	Flow in the pressure and return lines before the area ratio
Q_suc	65.25 L/min	Flow in the suction line
Q_prs	128.71 L/min	Flow in the line to the cap port (worst stroke)
Q_rtn	128.71 L/min	Flow in the return line

SYMBOL	VALUE	DESCRIPTION
v_suc	0.55386 m/s	Suction line velocity
v_prs	4.3702 m/s	Pressure line velocity (cap-port line, worst stroke)
v_rtn	2.6673 m/s	Return line velocity
u_suc	0.46155	Suction line velocity over its limit
u_prs	0.87403	Pressure line velocity over its limit
u_rtn	0.88911	Return line velocity over its limit
d_min_suc	33.969 mm	Smallest suction line bore
d_min_prs	23.372 mm	Smallest pressure line bore
d_min_rtn	30.174 mm	Smallest return line bore
d_rec_suc	38 mm	Recommended suction line bore
d_rec_prs	25 mm	Recommended pressure line bore
d_rec_rtn	32 mm	Recommended return line bore
L_eff	950 mm	Effective length
lam	42.222	Slenderness
lam_c	91.052	Transition slenderness
F_cr	2838.9 kN	Critical load (screening)
F_allow	811.11 kN	Allowable push load (screening)
F_stall	220.89 kN	Stall thrust p·A_b
F_buck	220.89 kN	Rod load judged for buckling: the larger of the push load and the stall thrust
u_buck	0.27234	Rod buckling utilisation (screening)
buckling_note	"screening only; ISO/TS 13725 method not implemented; stocky rod, Johnson parabola"	Rod buckling check

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Pump delivery covers the circuit flow	u_pump = 0.96794	u_pump ≤ 1	0.968	
PASS	Return line velocity within its limit	u_rtn = 0.88911	u_rtn ≤ 1	0.889	cylinder.line-limits record
PASS	Pressure line velocity within its limit, cap-port line at the larger stroke flow	u_prs = 0.87403	u_prs ≤ 1	0.874	cylinder.line-limits record
PASS	Extend force covers the push load	u_push = 0.7148	u_push ≤ 1	0.715	

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Retract force covers the pull load	u_pull = 0.59369	u_pull ≤ 1	0.594	
PASS	Suction line velocity within its limit	u_suc = 0.46155	u_suc ≤ 1	0.462	cylinder.line-limits record
PASS	Rod buckling at the larger of the push load and the stall thrust, screening only; ISO/TS 13725 method not implemented	u_buck = 0.27234	u_buck ≤ 1	0.272	Euler / Johnson column relations

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: E, f_y (Rod material); K_m (Mounting effective-length factor); S_f (Rod buckling safety factor); v_suc_max, v_prs_max, v_ret_max (Line velocity limits); hoses (Hose and pipe size list). Enter each as one so its source, edition and licence travel with it.
WARNING — input.eta_hm: eta_hm (Hydromechanical efficiency of the motor) was not entered; the template default 0.9 is used.
WARNING — input.C_d: C_d (Orifice discharge coefficient) was not entered; the template default 0.62 is used.

8. Verification

Template fp.cylinder@1.3.0 Content hash f94c2e869fa10d18 Engine calc-engine/1.0.0
Known-answer cases passed 7/7 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0